Section 1 Recorded!

① $X[B,D] \cdot Y[D,F] \rightarrow Z[B,F]$

a) $BD + DF$ bytes to load and BF to be written

b) $2BDF$

c) $AI = \dfrac{2BDF}{BD+DF+BF}$

d) $T_{math} = \dfrac{2BDF}{3.94e14}$ $T_{comms} = \dfrac{BD+DF+BF}{8.2e11}$

Reasonable upper bound : $T_{math} + T_{comms}$

Reasonable lower bound : $\max(T_{max}, T_{comms})$

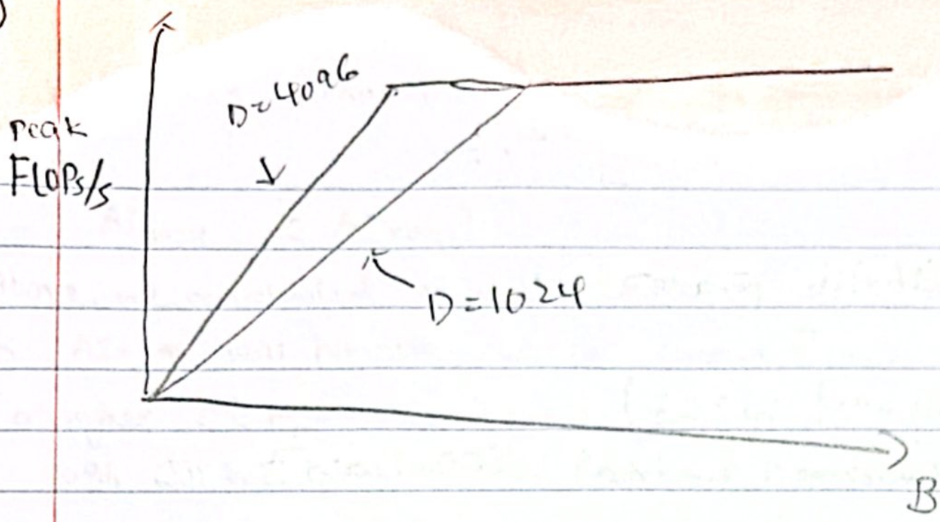
② $T_{math} = T_{comms}$ ($B > 240$ based on above notes)

Assume $B \ll D$ & $B \ll F$

Based on eqn (8) in notes, instead of $\frac{BDF}{DF}$, we ~~know~~ now have

$\frac{2BDF}{DF} = 2B > 240$ based on (8) $\Rightarrow B > 120$

3)



Total bytes = output & Activations + Weights

$= 2BD + 2BF + DF$

$T_{math} = T_{comms}$. Calculate at $B=100$

$$\cancel{\text{Intensity} = \frac{2BDF}{2BD+2BF+DF} = \frac{200 \cdot D^2}{200(2D)+D^2} = \frac{200\,D^2}{400\,D+D^2}}$$

$$\cancel{= \frac{200D}{400+D} \Rightarrow \text{At } D=4096 \quad \frac{200(4096)}{400+4096} =}$$

~~Assume $B \ll D$ & $B \ll F$,~~

$$\cancel{\text{Intensity} = 2BDF}$$

$$\text{Intensity} = \frac{2BDF}{2BD+2BF+DF} = \frac{2BD^2}{4BD+D^2} = \frac{2BD}{4B+D}$$

At $B=100$,

$$D=F=4096 = \frac{200 \cdot 4096}{400+4096} = 182$$

$$D=1024 = \frac{200 \cdot 1024}{400+1024} = 143$$

4) $AI = \frac{FLOPs}{comms}$

FLOPs : $[B, D]$ @ $[B, D, F]$ $= 2BDF$

#bytes : $BD + BDF + BF$

$AI = \frac{2BDF}{BD + BDF + BF}$ Assuming $B \ll D$, $B \ll F$, then

$AI = \frac{2BDF}{BDF} \approx 2$ which is constant

5) $B_{critical}$

Tensor core : 1979 tera FLOPs $\Rightarrow$ $\frac{1979}{2}$ tera FLOPs

BW : 3.35 terabytes

$B_{critical} = \frac{1979}{2} \cdot \frac{1}{3.35} = 295.37 \approx 295$ ✓